Introduction to Altair RapidMiner AI Studio:
How to Visually Design Data Analysis Workflows

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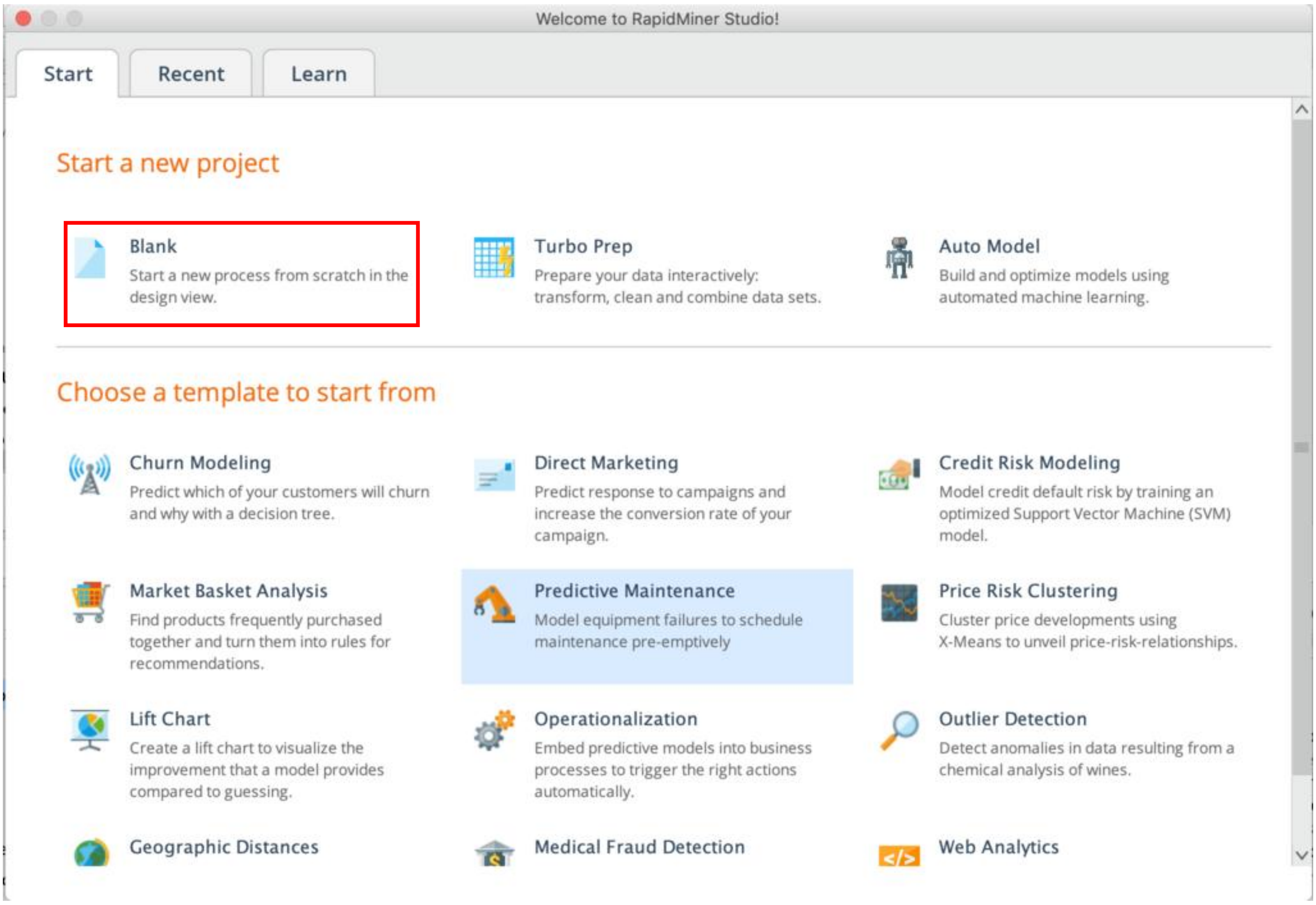
Altair RapidMiner is a software platform for data analytics, data science, machine learning (ML), and artificial intelligence (AI). Altair RapidMiner covers the complete data analysis process from data import from various types of data sources, data (pre-)processing, visualization, modelling with AI and ML, model validation, and model deployment. The platform offers several products including AI Studio (desktop), AI Hub (server), AI Real-Time Scoring Agents (RTSA) (edge), AI Cloud (cloud), and other products. This technical report focuses on Altair RapidMiner AI Studio and on how to use it to visually design and run data analysis workflows, which are also called RapidMiner processes.

Altair RapidMiner AI Studio is a software environment for creating and running data analysis and machine learning workflows, which are also called processes. AI Studio allows you to perform interactive data analysis and visualization as well as to create data analysis processes by combining predefined building blocks (operators) through a drag-and-drop graphical user interface (GUI) and to run these processes and to inspect their results. Instead of writing code you work visually by placing operators and setting their parameters in an easy-to-use graphical user interface (GUI).

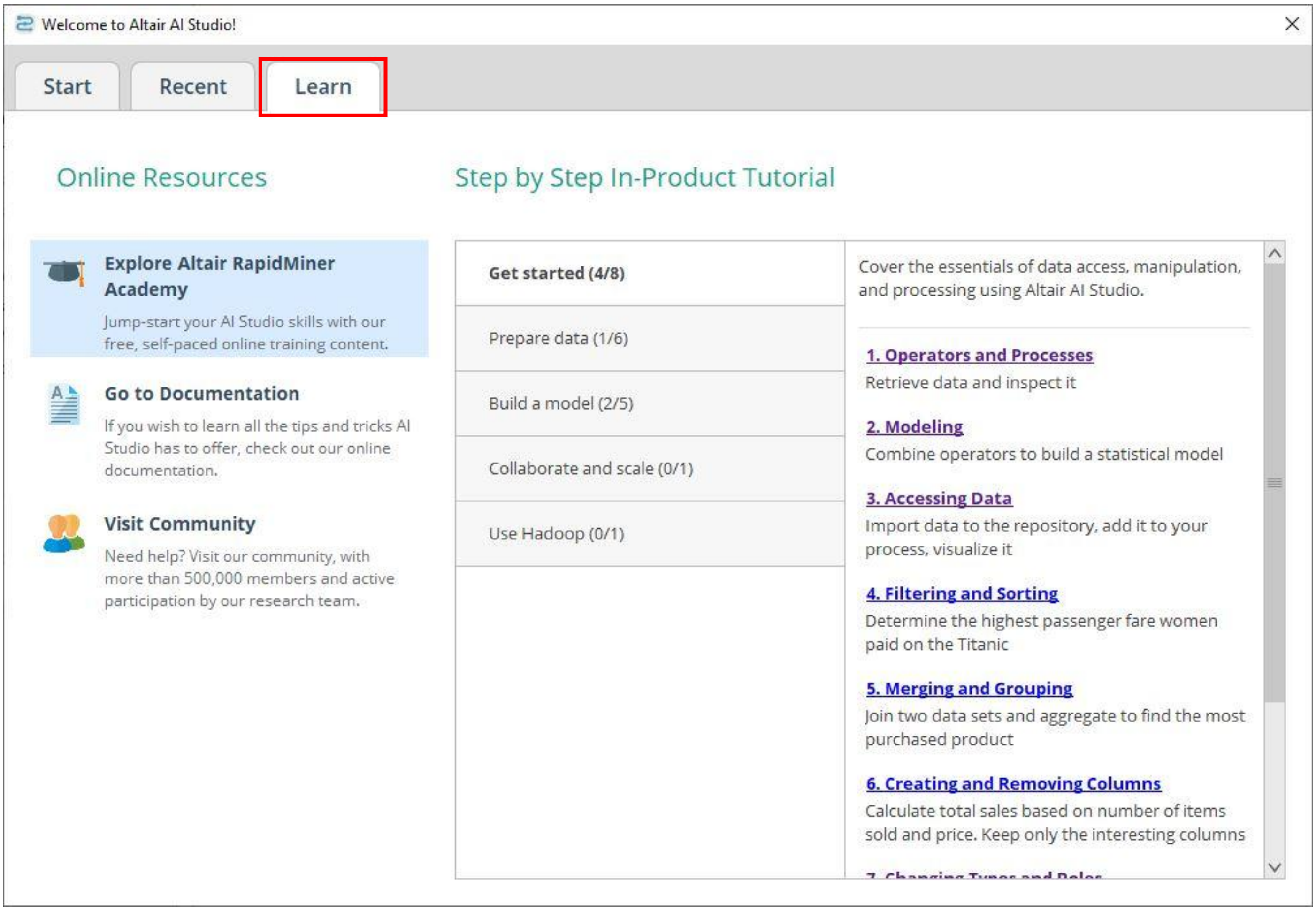
Here is a link to help you install AI Studio on your computer:
[Installing Altair AI Studio - Altair RapidMiner Documentation](https://docs.rapidminer.com/latest/studio/installation/index.html)
https://docs.rapidminer.com/latest/studio/installation/index.html

After installing and starting AI Studio, you have the option to start with a blank process. Click on this option to see the default view.

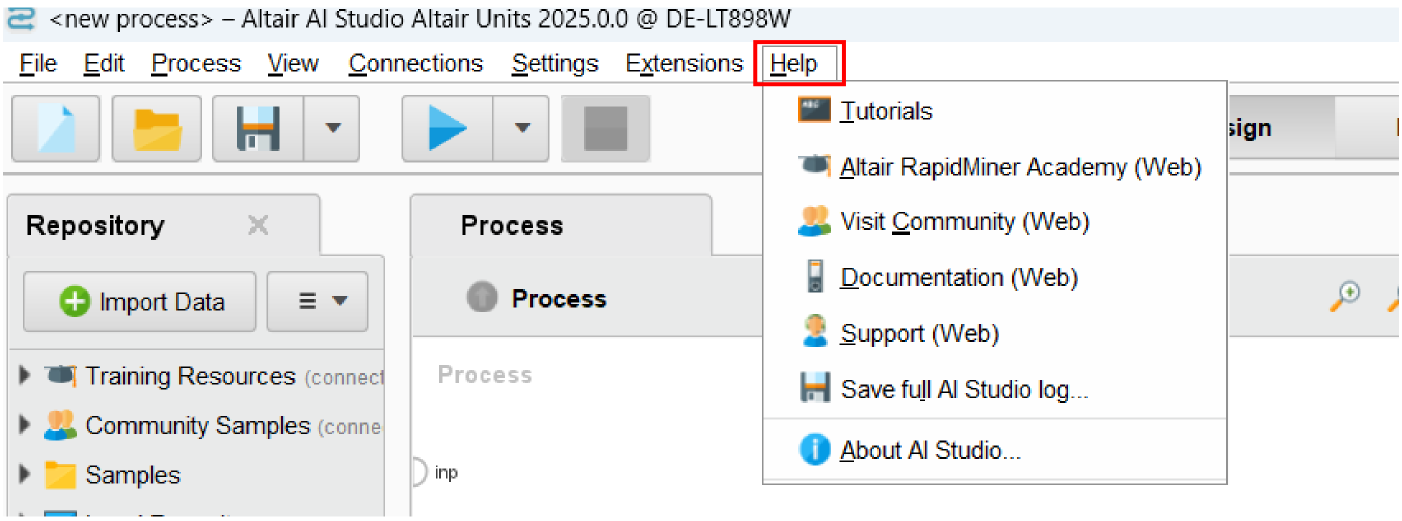
If you would like more help with your data preparation and model building, you can use "Turbo Prep" or "Auto Model" to accelerate the process. Additionally, you can select one of several different example templates as your starting point for designing your own data analysis workflow:



To get started, go to the learn tab on the welcome page. Here you will find step-by-step tutorials that will guide you through the different stages of building your own data analysis process:



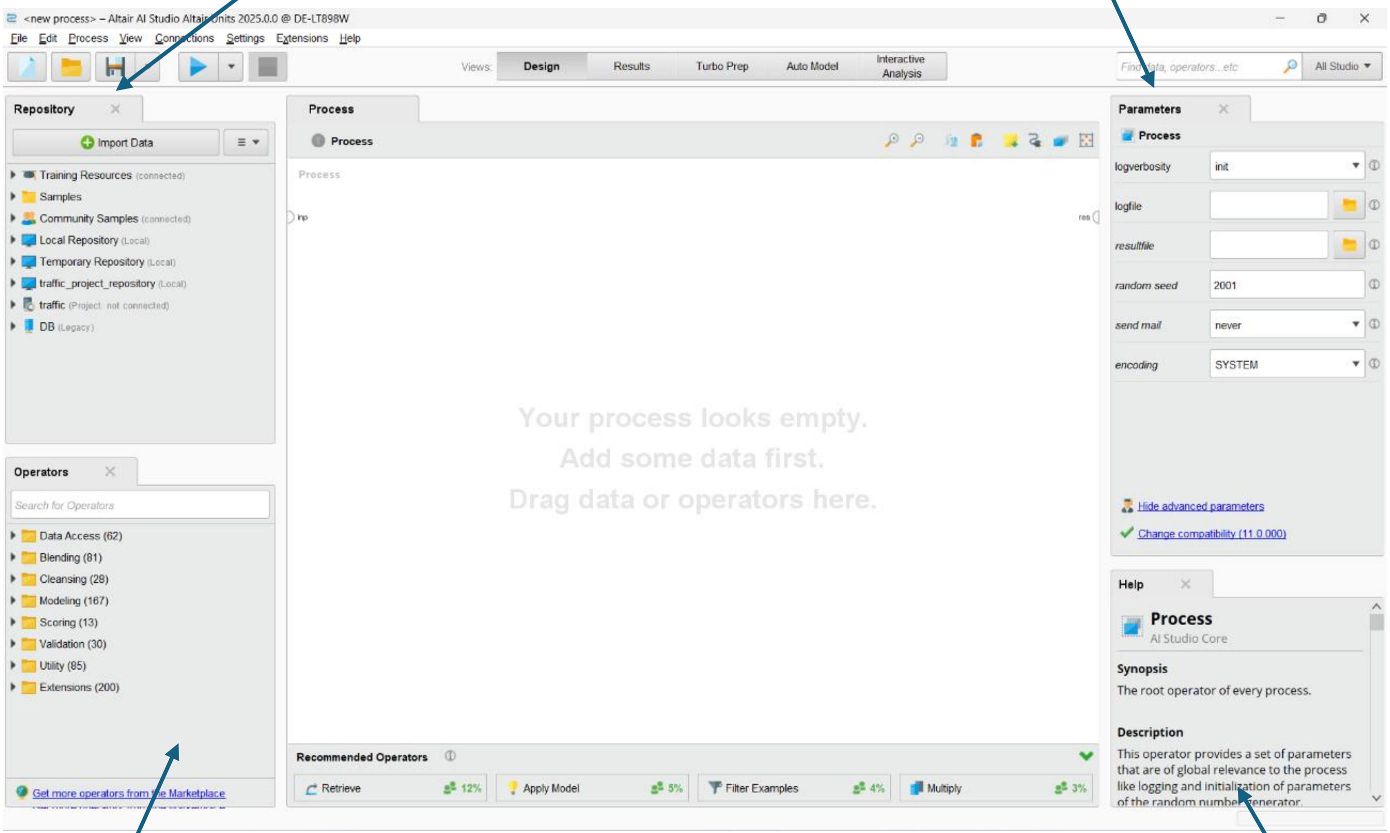
Later you can reopen the tutorials by selecting "Help" and then "Tutorials" in the top left corner of the default view of AI Studio:



Below is the default view of AI Studio where the different panels are displayed:

Data, processes, models, and results are stored in a repository

The behaviour of an operator can be modified by changing its parameters



The essential elements of every workflow are called operators

The behaviour of an operator can be understood by reading the help text

For a more detailed description of the AI Studio panels and their functions please visit:

[The Design View - Altair RapidMiner Documentation](https://docs.rapidminer.com/latest/studio/getting-started/design-view.html)
https://docs.rapidminer.com/latest/studio/getting-started/design-view.html

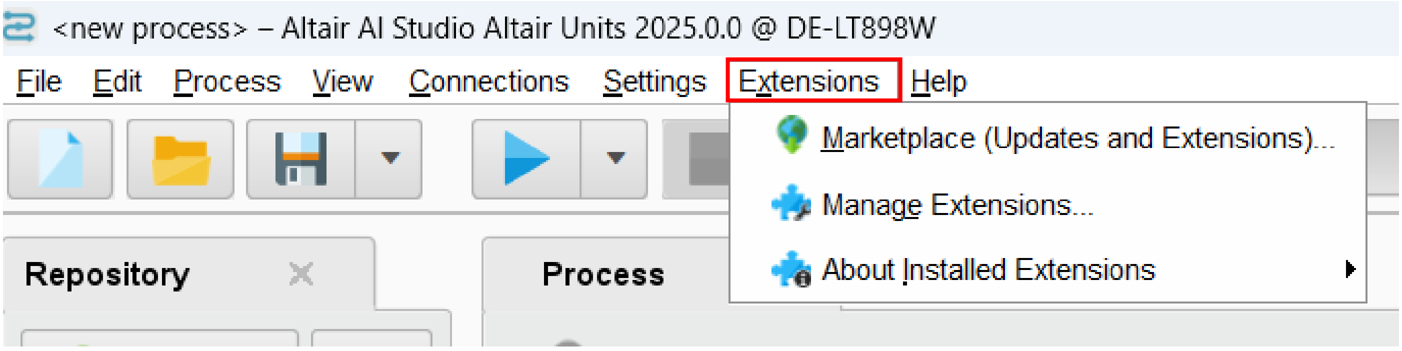
For a video introduction to AI Studio please visit:

[What is what introduction for RapidMiner | RapidMiner Studio](https://academy.rapidminer.com/learn/video/rapidminer-studio-gui-intro)
https://academy.rapidminer.com/learn/video/rapidminer-studio-gui-intro

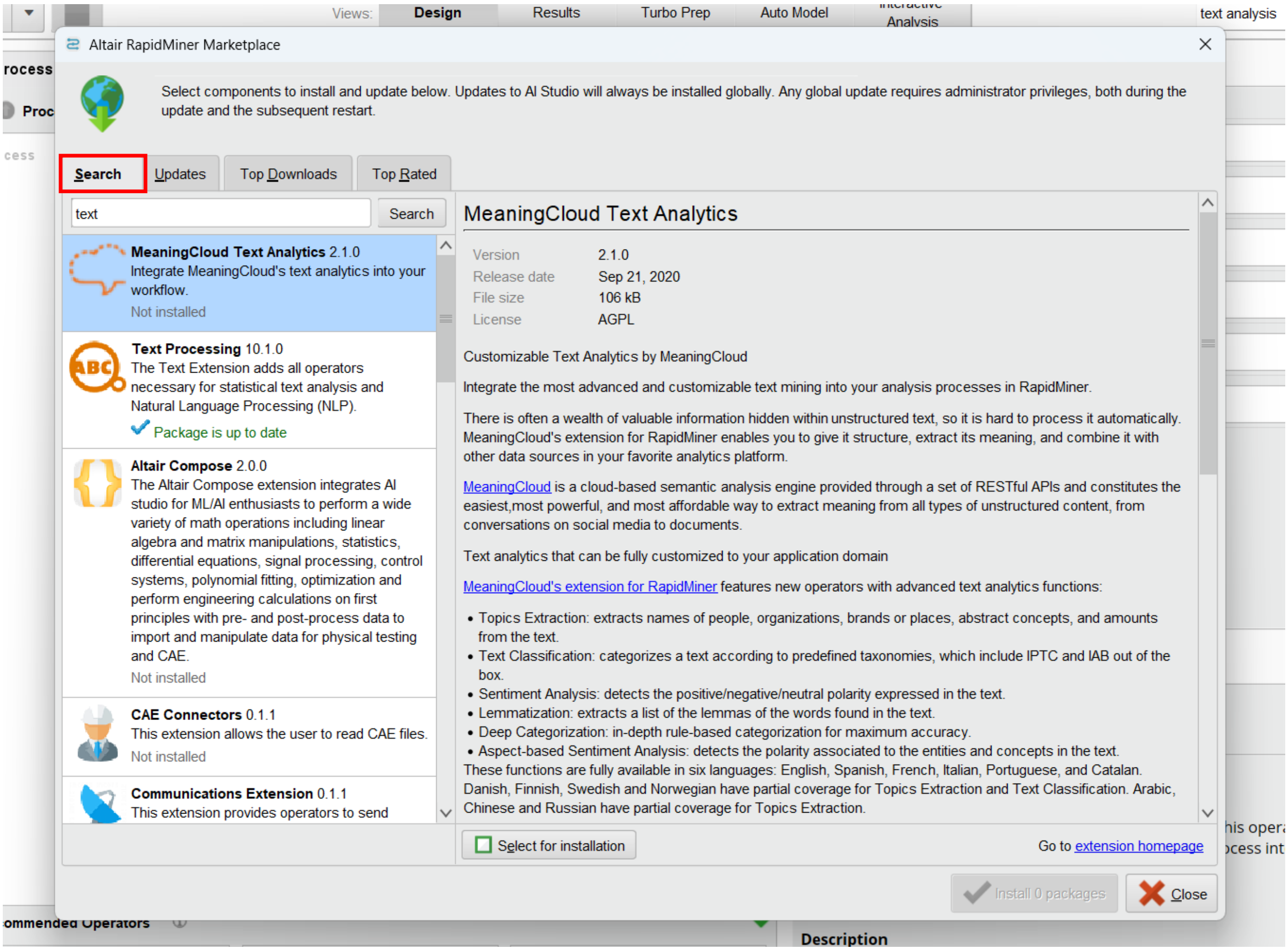
To build a process you need to find fitting operators for your intended result and add them to your data analysis process via drag and drop. To find fitting operatos, you can use the global search in the top right corner of the AI Studio GUI. With the global search, you can also find data and processes from the repository and actions you can take from the menu:

If you would like to enhance your data projects with, for example, text analysis, image analysis, or advanced data visualization, you can download extensions for AI Studio from the Altair RapidMiner Marketplace. These extensions can also be found using the global search within AI Studio. You can visit the Marketplace at [RapidMiner Marketplace](https://marketplace.rapidminer.com/) (https://marketplace.rapidminer.com/).

Additionally, you can download the extensions from within AI Studio by selecting "Extensions" and then "Marketplace" directly in AI Studio:



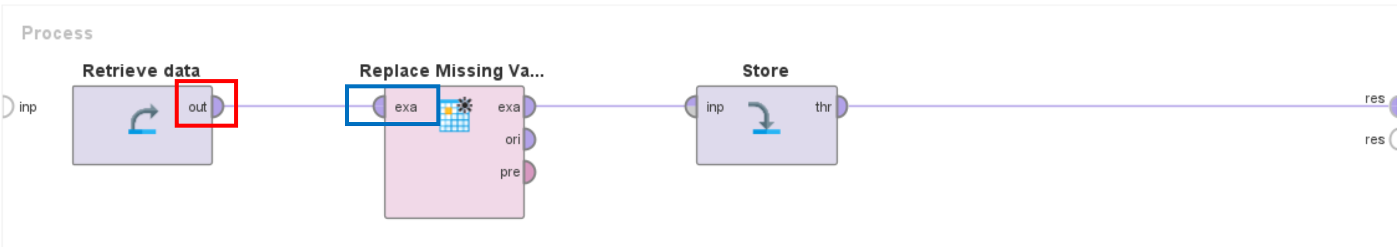
Under “Search”, you enter the name of the extension you are looking for or some search terms like for example “text” or “image” to describe the functionality you are looking for, and click on “select for installation” and then on “install packages”:



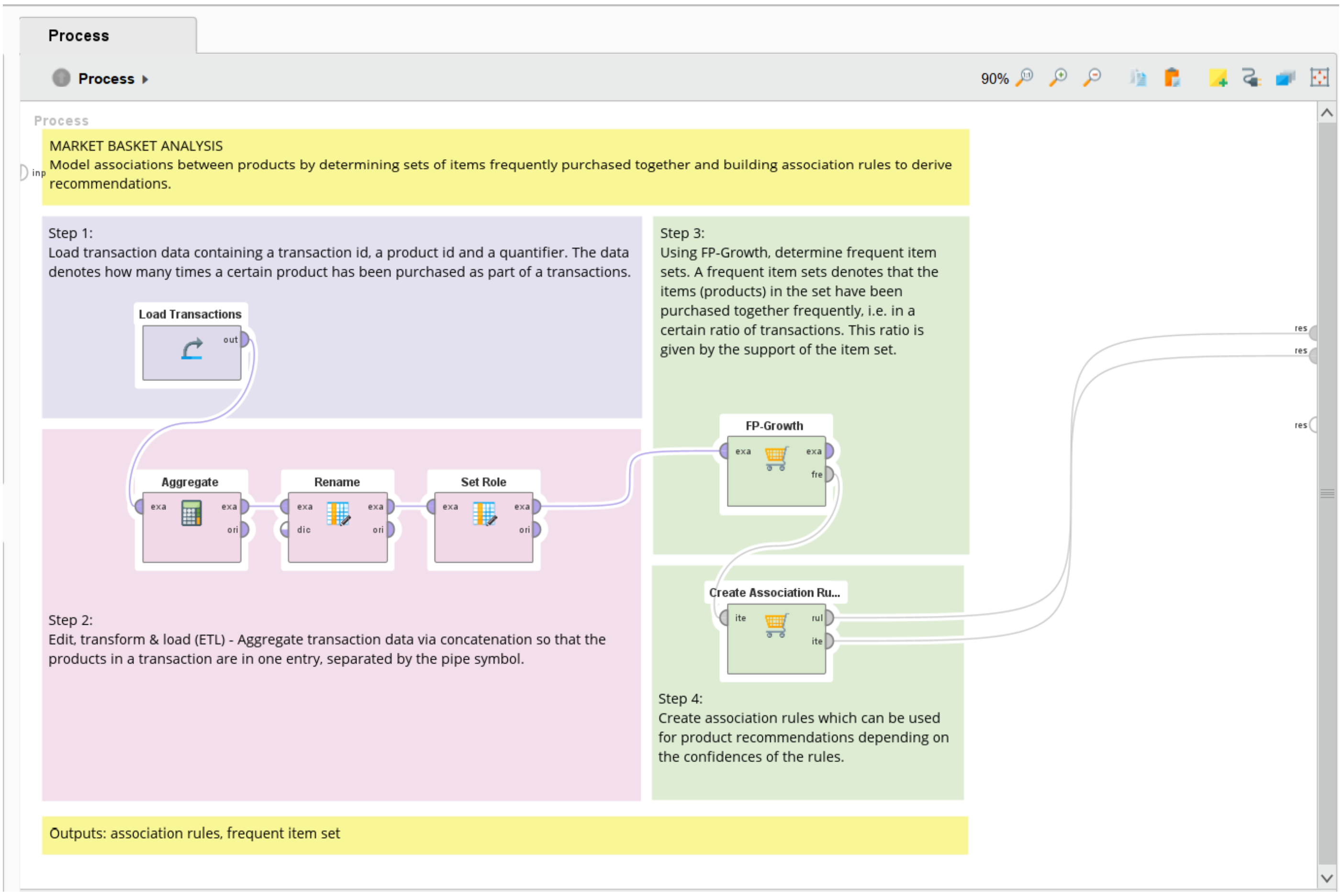
To complete your data analysis process, you need to connect the operators in your data analysis process by clicking on the output port of the first operator and then on the input port of the second operator and so. Data streams will follow these connections when the process runs. Make sure you only connect compatible ports to avoid an error message. A complete list and description of all ports can be found here:

[Getting Started glossary - Altair RapidMiner Documentation](https://docs.rapidminer.com/latest/studio/getting-started/important-terms.html#port-info)

https://docs.rapidminer.com/latest/studio/getting-started/important-terms.html#port-info



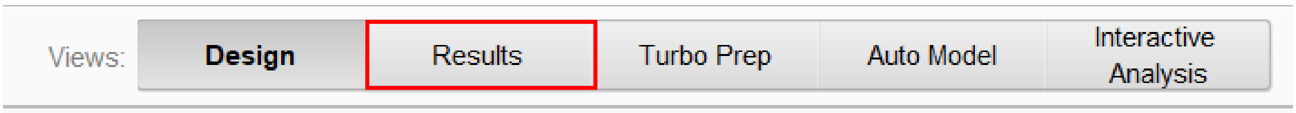
This is an example of a process that performs a market basket analysis. Read through the four steps to get an idea of what the different operators are used for:



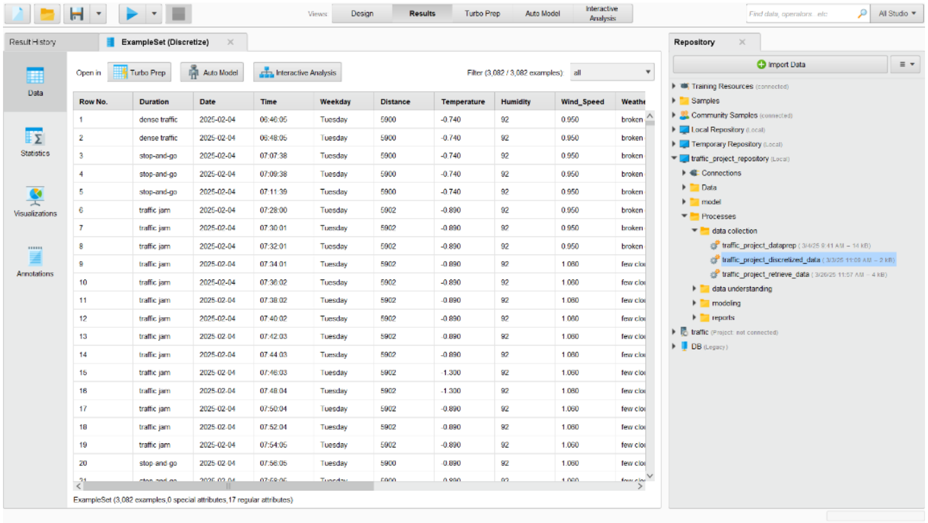
After designing your data analysis workflow and setting the parameters of the operators in your workflow, press the run button in the upper left corner to run the process.

There are several ways to run your process: locally on your computer, in the background, or on an Altair RapidMiner AI Hub (server). Click here for more information:
[Run a process - Altair RapidMiner Documentation](https://docs.rapidminer.com/latest/studio/getting-started/run-a-process.html)
https://docs.rapidminer.com/latest/studio/getting-started/run-a-process.html

After executing your process, the results will be displayed in the Results view.



Here you can see the resulting data and statistics. You can also click on the visualizations tab to generate plots and graphs.



For more tutorials about Altair RapidMiner AI Studio please visit the Altair RapidMiner Academy:
[Machine Learning and RapidMiner Tutorials | Altair Engineering Inc. Academy](https://academy.rapidminer.com/)
https://academy.rapidminer.com/

For academic use of Altair RapidMiner AI Studio and for resources for getting started like free online courses, tutorial videos, use case examples, books, etc. please visit:
[Altair RapidMiner AI Studio Academic Hub](https://web.altair.com/academic-hub-ai-studio/)
https://web.altair.com/academic-hub-ai-studio/

For help, best practices, and networking please visit the Altair RapidMiner Community:
[Altair Community](https://community.altair.com/)
https://community.altair.com/

For more information about the Altair RapidMiner platform please visit
[Altair RapidMiner](https://altair.com/altair-rapidminer)
https://altair.com/altair-rapidminer

If you have further questions about AI Studio or the project, please contact crexdata@altair.com .